

# Chapter 4 Homework: Prediction, Goodness-of-Fit, and Modeling Issues

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## Exercise 4.1

Answer each of the following:

### Part (a)

Suppose that a simple regression has quantities  $N = 20$ ,  $\sum y_i^2 = 7825.94$ ,  $\bar{y} = 19.21$ , and  $SSR = 375.47$ . Find  $R^2$ .

```
N      <- 20
sum_y2 <- 7825.94
ybar    <- 19.21
SSR     <- 375.47

# SST = sum(yi^2) - N * ybar^2
SST <- sum_y2 - N * ybar^2

# R^2 = 1 - SSR/SST (here SSR = sum of squared residuals = SSE)
R2 <- 1 - SSR / SST

cat("SST =", round(SST, 4), "\n")
```

```
## SST = 445.458
```

```
cat("R^2 =", round(R2, 4), "\n")
```

```
## R^2 = 0.1571
```

**Interpretation:**  $R^2 \approx 0.1571$ , meaning about 15.71% of the variation in  $y$  is explained by the regression.

### Part (b)

Suppose that a simple regression has quantities  $R^2 = 0.7911$ ,  $SST = 725.94$ , and  $N = 20$ . Find  $\sigma^2$ .

```
R2b <- 0.7911
SST2 <- 725.94
N2 <- 20

# SSE = (1 - R^2) * SST
SSE <- (1 - R2b) * SST2

# sigma_hat^2 = SSE / (N - 2)
sigma2_hat <- SSE / (N2 - 2)

cat("SSE =", round(SSE, 4), "\n")
```

```
## SSE = 151.6489
```

```
cat("sigma_hat^2 =", round(sigma2_hat, 4), "\n")
```

```
## sigma_hat^2 = 8.4249
```

**Interpretation:** The estimated error variance  $\sigma^2 \approx 8.4249$ .

## Part (c)

Suppose that a simple regression has quantities  $\sum (y_i - \bar{y})^2 = 631.63$  and  $\sum e_i^2 = 182.85$ . Find  $R^2$ .

```
SST3 <- 631.63
SSE3 <- 182.85

R2c <- 1 - SSE3 / SST3

cat("R^2 =", round(R2c, 4), "\n")
```

```
## R^2 = 0.7105
```

**Interpretation:**  $R^2 \approx 0.7105$ .

## Exercise 4.3

We have five observations:  $x_i = (3, 2, 1, -1, 0)$  and  $y_i = (4, 2, 3, 1, 0)$ . The fitted least squares line is  $\hat{y}_i = 1.2 + 0.8x_i$ ,  $\sum_{i=1}^5 e_i^2 = 3.6$ ,  $\sum (x_i - \bar{x})^2 = 10$ ,  $\sum (y_i - \bar{y})^2 = 10$ .

```
x    <- c(3, 2, 1, -1, 0)
y    <- c(4, 2, 3, 1, 0)
N    <- length(x)
b1   <- 1.2; b2 <- 0.8
SSE  <- 3.6
SSx  <- 10    # sum((xi - xbar)^2)

xbar <- mean(x)
ybar <- mean(y)

sigma2 <- SSE / (N - 2)  # estimated error variance
sigma  <- sqrt(sigma2)

cat("x-bar:", xbar, " | y-bar:", ybar, "\n")
```

```
## x-bar: 1 | y-bar: 2
```

```
cat("sigma^2:", sigma2, " | sigma:", round(sigma, 4), "\n")
```

```
## sigma^2: 1.2 | sigma: 1.0954
```

## Part (a): Predicted value at $x_0 = 4$

```
x0      <- 4
y_hat0 <- b1 + b2 * x0
cat("Predicted y at x0 =", x0, ":", y_hat0, "\n")
```

```
## Predicted y at x0 = 4 : 4.4
```

## Part (b): Standard error of forecast $se(f)$

$$se(f) = \hat{\sigma} \sqrt{1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

```
se_f <- sigma * sqrt(1 + 1/N + (x0 - xbar)^2 / SSx)
cat("se(f) =", round(se_f, 4), "\n")
```

```
## se(f) = 1.5875
```

## Part (c): 95% prediction interval

```
alpha <- 0.05
t_crit_95 <- qt(1 - alpha/2, df = N - 2)

PI_95_lower <- y_hat0 - t_crit_95 * se_f
PI_95_upper <- y_hat0 + t_crit_95 * se_f

cat("t critical (95%, df=3):", round(t_crit_95, 4), "\n")
```

```
## t critical (95%, df=3): 3.1824
```

```
cat("95% Prediction Interval: [", round(PI_95_lower, 4), ",", round(PI_95_upper, 4),
    "]\n")
```

```
## 95% Prediction Interval: [ -0.652 , 9.452 ]
```

## Part (d): 99% prediction interval

```
t_crit_99 <- qt(0.995, df = N - 2)

PI_99_lower <- y_hat0 - t_crit_99 * se_f
PI_99_upper <- y_hat0 + t_crit_99 * se_f

cat("t critical (99%, df=3):", round(t_crit_99, 4), "\n")
```

```
## t critical (99%, df=3): 5.8409
```

```
cat("99% Prediction Interval: [", round(PI_99_lower, 4), ",", round(PI_99_upper, 4),
    "]\n")
```

```
## 99% Prediction Interval: [ -4.8722 , 13.6722 ]
```

## Part (e): 95% prediction interval at $x_0 = \bar{x}$

```
x0_bar <- xbar
y_hat_xbar <- b1 + b2 * x0_bar
se_f_xbar <- sigma * sqrt(1 + 1/N + (x0_bar - xbar)^2 / SSx)

PI_xbar_lower <- y_hat_xbar - t_crit_95 * se_f_xbar
PI_xbar_upper <- y_hat_xbar + t_crit_95 * se_f_xbar

cat("Prediction at xbar:", y_hat_xbar, "\n")
```

```
## Prediction at xbar: 2
```

```
cat("se(f) at xbar:", round(se_f_xbar, 4), "\n")
```

```
## se(f) at xbar: 1.2
```

```
cat("95% PI at xbar: [", round(PI_xbar_lower, 4), ",", round(PI_xbar_upper, 4), "]\n")
```

```
## 95% PI at xbar: [ -1.8189 , 5.8189 ]
```

```
cat("Width at x0=xbar:", round(PI_xbar_upper - PI_xbar_lower, 4), "\n")
```

```
## Width at x0=xbar: 7.6379
```

```
cat("Width at x0=4:  ", round(PI_95_upper - PI_95_lower, 4), "\n")
```

```
## Width at x0=4:    10.104
```

**Interpretation:** The interval at  $x$  is narrower than at  $x_0 = 4$  because prediction uncertainty increases as  $x_0$  moves away from  $x$ .

# Exercise 4.25

The file `collegetown` contains data on 500 houses sold in Baton Rouge, LA during 2009–2013.

```
# Load data
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/collegetown.rdata"))
head(collegetown)
```

|        | price<br><dbl> | sqft<br><dbl> | age<br><int> | pool<br><int> | fireplace<br><int> | close<br><int> | twostory<br><int> | occupied<br><int> |
|--------|----------------|---------------|--------------|---------------|--------------------|----------------|-------------------|-------------------|
| 1      | 96.5           | 16.77         | 10           | 0             | 0                  | 0              | 0                 | 0                 |
| 2      | 106.0          | 17.51         | 10           | 0             | 0                  | 0              | 0                 | 0                 |
| 3      | 114.0          | 18.37         | 9            | 0             | 0                  | 0              | 0                 | 1                 |
| 4      | 119.9          | 29.10         | 9            | 0             | 0                  | 0              | 0                 | 0                 |
| 5      | 122.7          | 14.68         | 9            | 0             | 0                  | 0              | 0                 | 1                 |
| 6      | 125.0          | 18.23         | 8            | 0             | 0                  | 0              | 0                 | 0                 |
| 6 rows |                |               |              |               |                    |                |                   |                   |

## Part (a): Log-linear model

$$\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$$

```
collegetown$lnPRICE <- log(collegetown$price)
collegetown$lnSQFT  <- log(collegetown$sqft)

# Log-linear model
model_ll <- lm(lnPRICE ~ sqft, data = collegetown)
summary(model_ll)
```

```
##
## Call:
## lm(formula = lnPRICE ~ sqft, data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.32528 -0.19199  0.00122  0.21678  0.86609
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.393866   0.043288  101.50  <2e-16 ***
## sqft         0.036044   0.001486   24.26  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.34 on 498 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5408
## F-statistic: 588.7 on 1 and 498 DF, p-value: < 2.2e-16
```

```
# Slope and elasticity at sample means
b1_ll <- coef(model_ll)[1]
b2_ll <- coef(model_ll)[2]
sqft_mean <- mean(collegetown$sqft)
price_mean <- mean(collegetown$price)

slope_ll <- b2_ll * exp(b1_ll + b2_ll * sqft_mean)
elasticity_ll <- b2_ll * sqft_mean

cat("\nSlope at mean SQFT:", round(slope_ll, 4), "\n")
```

```
##
## Slope at mean SQFT: 7.8009
```

```
cat("Elasticity at mean SQFT:", round(elasticity_ll, 4), "\n")
```

```
## Elasticity at mean SQFT: 0.9834
```

**Interpretation:** In the log-linear model,  $\hat{\beta}_2$  represents the approximate percentage change in PRICE for a one-unit increase in SQFT. The slope at the mean is 7.8.

## Part (b): Log-log model

$$\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$$

```
model_llog <- lm(lnPRICE ~ lnSQFT, data = collegetown)
summary(model_llog)
```

```
##
## Call:
## lm(formula = lnPRICE ~ lnSQFT, data = collegetown)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.36864 -0.20849  0.00487  0.23204  1.21099
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.04965     0.15797   12.97  <2e-16 ***
## lnSQFT       1.02483     0.04839   21.18  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3643 on 498 degrees of freedom
## Multiple R-squared:  0.4738, Adjusted R-squared:  0.4728
## F-statistic: 448.5 on 1 and 498 DF,  p-value: < 2.2e-16
```

```
a1 <- coef(model_llog)[1]
a2 <- coef(model_llog)[2]

slope_llog <- a2 * exp(a1 + a2 * log(sqft_mean)) / sqft_mean
elasticity_llog <- a2 # constant elasticity in log-log

cat("\nSlope at mean SQFT:", round(slope_llog, 4), "\n")
```

```
##
## Slope at mean SQFT: 8.6388
```

```
cat("Elasticity (constant):", round(elasticity_llog, 4), "\n")
```

```
## Elasticity (constant): 1.0248
```

**Interpretation:** In the log-log model,  $\hat{\alpha}_2$  is the constant price elasticity with respect to SQFT.

## Part (c): Compare $R^2$ for linear, log-linear, and

# log-log models

```
# Linear model
model_lin <- lm(price ~ sqft, data = collegetown)

# R^2 for linear
R2_lin <- summary(model_lin)$r.squared

# Generalized R^2 for log-linear (squared correlation between actual PRICE and corrected predictor)
sigma2_ll <- summary(model_ll)$sigma^2
yhat_c_ll <- exp(fitted(model_ll)) * exp(sigma2_ll / 2)
gR2_ll <- cor(collegetown$price, yhat_c_ll)^2

# Generalized R^2 for log-log
sigma2_llog <- summary(model_llog)$sigma^2
yhat_c_llog <- exp(fitted(model_llog)) * exp(sigma2_llog / 2)
gR2_llog <- cor(collegetown$price, yhat_c_llog)^2

cat("Linear model R^2:          ", round(R2_lin, 4), "\n")
```

```
## Linear model R^2:          0.6413
```

```
cat("Log-linear generalized R^2:", round(gR2_ll, 4), "\n")
```

```
## Log-linear generalized R^2: 0.6622
```

```
cat("Log-log generalized R^2:   ", round(gR2_llog, 4), "\n")
```

```
## Log-log generalized R^2:    0.6445
```

## Part (d): Jarque–Bera normality tests on residuals

```
library(tseries)

jb_lin <- jarque.bera.test(residuals(model_lin))
jb_ll <- jarque.bera.test(residuals(model_ll))
jb_llog <- jarque.bera.test(residuals(model_llog))

cat("Jarque–Bera test p-values:\n")
```

```
## Jarque–Bera test p-values:
```

```
cat("Linear:          ", round(jb_lin$p.value, 4), "\n")
```

```
## Linear:          0
```



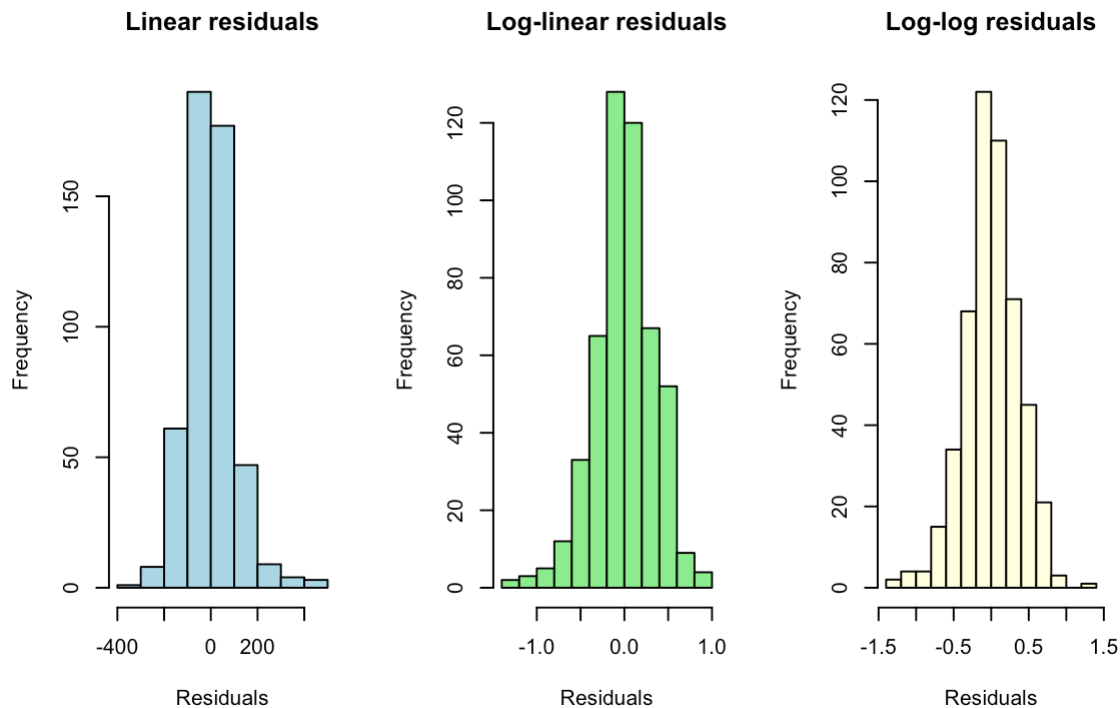
```
cat("Log-linear: ", round(jb_ll$p.value, 4), "\n")
```

```
## Log-linear: 0
```

```
cat("Log-log: ", round(jb_lllog$p.value, 4), "\n")
```

```
## Log-log: 8e-04
```

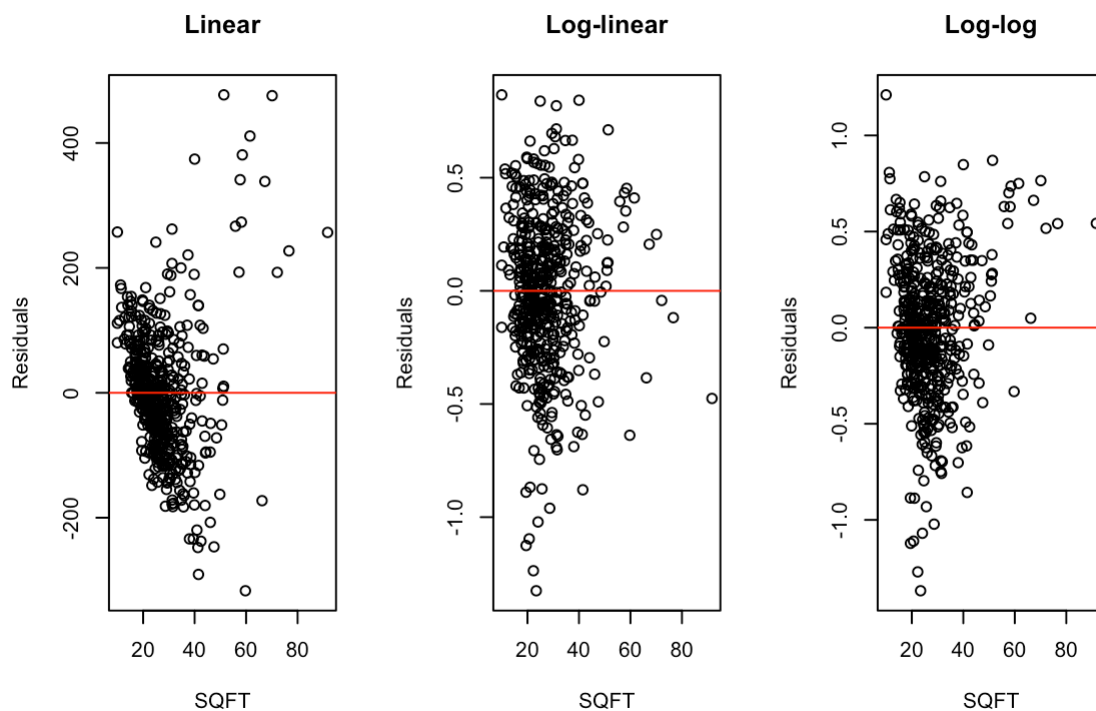
```
# Histograms
par(mfrow = c(1, 3))
hist(residuals(model_lin), main = "Linear residuals", xlab = "Residuals", col =
"lightblue")
hist(residuals(model_ll), main = "Log-linear residuals", xlab = "Residuals", col =
"lightgreen")
hist(residuals(model_lllog), main = "Log-log residuals", xlab = "Residuals", col =
"lightyellow")
```



```
par(mfrow = c(1, 1))
```

## Part (e): Residuals vs. SQFT

```
par(mfrow = c(1, 3))
plot(collegetown$sqft, residuals(model_lin), main = "Linear", xlab = "SQFT", ylab = "Residuals")
abline(h = 0, col = "red")
plot(collegetown$sqft, residuals(model_ll), main = "Log-linear", xlab = "SQFT", ylab = "Residuals")
abline(h = 0, col = "red")
plot(collegetown$sqft, residuals(model_lllog), main = "Log-log", xlab = "SQFT", ylab = "Residuals")
abline(h = 0, col = "red")
```



```
par(mfrow = c(1, 1))
```

## Part (f): Point predictions for a house with 2700

# sq ft

```
x_new <- 2700

# Linear
pred_lin <- predict(model_lin, newdata = data.frame(sqft = x_new))

# Log-linear (corrected predictor)
pred_ll_ln <- predict(model_ll, newdata = data.frame(sqft = x_new))
pred_ll    <- exp(pred_ll_ln) * exp(sigma2_ll / 2)

# Log-log (corrected predictor)
pred_llog_ln <- predict(model_llog, newdata = data.frame(lnSQFT = log(x_new)))
pred_llog    <- exp(pred_llog_ln) * exp(sigma2_llog / 2)

cat("Point predictions for SQFT = 2700:\n")
```

```
## Point predictions for SQFT = 2700:
```

```
cat("Linear:      $", round(pred_lin, 2), "thousand\n")
```

```
## Linear:      $ 36072.5 thousand
```

```
cat("Log-linear: $", round(pred_ll, 2), "thousand\n")
```

```
## Log-linear: $ 1.580978e+44 thousand
```

```
cat("Log-log:     $", round(pred_llog, 2), "thousand\n")
```

```
## Log-log:     $ 27260.46 thousand
```

## Part (g): 95% prediction intervals

```
# Linear PI
pi_lin <- predict(model_lin, newdata = data.frame(sqft = x_new),
              interval = "prediction", level = 0.95)

# Log-linear PI (transform back)
pi_ll_ln <- predict(model_ll, newdata = data.frame(sqft = x_new),
              interval = "prediction", level = 0.95)
pi_ll <- exp(pi_ll_ln)

# Log-log PI (transform back)
pi_llog_ln <- predict(model_llog, newdata = data.frame(lnSQFT = log(x_new)),
              interval = "prediction", level = 0.95)
pi_llog <- exp(pi_llog_ln)

cat("95% Prediction Intervals for SQFT = 2700:\n")
```

```
## 95% Prediction Intervals for SQFT = 2700:
```

```
cat("Linear:      [", round(pi_lin[2], 2), ",", round(pi_lin[3], 2), "]\n")
```

```
## Linear:      [ 33705.21 , 38439.8 ]
```

```
cat("Log-linear: [", round(pi_ll[2], 2), ",", round(pi_ll[3], 2), "]\n")
```

```
## Log-linear: [ 5.935131e+40 , 3.751584e+47 ]
```

```
cat("Log-log:     [", round(pi_llog[2], 2), ",", round(pi_llog[3], 2), "]\n")
```

```
## Log-log:     [ 10989.49 , 59216.84 ]
```

## Part (h): Discussion of functional form choice

Based on the generalized  $R^2$  comparisons, residual patterns, and Jarque–Bera tests, the **log-log** or **log-linear** models generally outperform the linear specification for house price data:

- The linear residuals often fan out with SQFT, suggesting heteroskedasticity.
- The log transformations produce more homogeneous residuals.
- The generalized  $R^2$  values suggest better fit for the log models.
- The log-log model has the advantage of a constant elasticity interpretation.